Linear systems and elimination

A sharp subset bound for worst-case normal GMRES

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging reflects extending a discrete real approximation principle to complex spectra with a sharp constant; community impact concerns usable worst-case GMRES bounds for normal matrices.

Let $L \subset \mathbb{C} \setminus \{0\}$ consist of $n \geq 3$ distinct points. For a nonempty $S \subseteq L$, define

$$M_k(S) = \min_{p \in \mathbb{C}[z],\ \deg p \leq k,\ p(0)=1} \max_{z \in S} |p(z)|.$$

Is it true, simultaneously for every such L and $1 \leq k \leq n-2$, that

$$M_k(L) \leq \frac{4}{\pi} \max_{S \subseteq L,\ |S|=k+1} M_k(S)?$$

The constant is required to be independent of dimension, degree, and the locations of the points.

For a normal matrix with spectrum L, $M_k(L)$ is the largest relative residual norm attainable at GMRES step k, over unit initial residuals. The smaller-set quantities have explicit formulas through Lagrange interpolation. The question therefore asks for a sharp, dimension-independent certificate of worst-case convergence using small subsets of the spectrum. It is distinct from equality of ideal and worst-case GMRES for a nonnormal Jordan block (IE-02).

References

J. Liesen and P. Tichý, *The worst-case GMRES for normal matrices*, BIT 44 (2004), 79–98, conjecture (3.16), pp. 91–92, and Appendix (author copy). Their *A min-max problem on roots of unity*, TU Berlin Preprint 28-2003, conclusion (9.1), proves selected roots-of-unity cases and sharpness of $4/\pi$ (university record). Their survey *Convergence analysis of Krylov subspace methods*, GAMM-Mitteilungen 27 (2004), discussion after (11), restates the conjecture (author copy).

Status check — 2026-09-08

Searched the original title with "conjecture", "4/pi", "4/π", "proof", "counterexample", and 2025–2026; checked the authors' current publication lists. The later Jordan-block papers concern a different conjecture. No resolution or recent explicit reaffirmation of this particular bound was located; its open-status evidence is historical and bounded.

Audit update — 2026-09-10

The primary paper, §3.2.1, proves the stronger subset equality for every real spectrum, a substantive subclass of the displayed target. Its complex-spectrum conjecture remains in §3.2.2. Searches for the Liesen–Tichý conjecture and later sharp normal-GMRES subset bounds found no general resolution; the current status records the real-spectrum partial result without asserting exhaustive literature coverage.